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Full Stack Development — Mini E-Commerce Project Lab Sheet 2: Making It Interactive (Vanilla JavaScript)

PART: 01 JAVASCRIPT (script.js)

Product catalog, cart logic (add/remove/update), dynamic rendering, calculateTotal(), form validation and the cart-badge updater are all implemented in a single linked file, script.js.

```
/* =====  
TechMart - script.js  
Lab Sheet 2: Cart logic, dynamic rendering, form validation  
===== */  
  
/* ----- 1. Product catalog ----- */  
const products = [  
  { id: 1, name: "Wireless Earbuds", price: 1999, image: "earbuds_real.png" },  
  { id: 2, name: "Smart Watch", price: 3499, image: "watch_real.png" },  
  { id: 3, name: "Bluetooth Speaker", price: 1299, image: "speaker_real.png" },  
  { id: 4, name: "Power Bank", price: 999, image: "powerbank_real.png" },  
  { id: 5, name: "Mechanical Keyboard", price: 2799, image: "keyboard_real.png" },  
  { id: 6, name: "Wireless Mouse", price: 699, image: "mouse_real.png" },  
  { id: 7, name: "HD Webcam", price: 1599, image: "webcam_real.png" },  
  { id: 8, name: "Laptop Stand", price: 899, image: "standd_real.png" }  
];  
  
/* ----- Cart helpers (localStorage) ----- */  
let cart = JSON.parse(localStorage.getItem("cart")) || [];  
  
function saveCart() {  
  localStorage.setItem("cart", JSON.stringify(cart));  
  updateCartBadge();  
}  
  
function addToCart(productId) {  
  const product = products.find((p) => p.id === productId);  
  if (!product) return;  
  
  const existingItem = cart.find((item) => item.id === productId);  
  if (existingItem) {  
    existingItem.quantity += 1;  
  } else {  
    cart.push({ ...product, quantity: 1 });  
  }  
  saveCart();  
}  
  
function removeFromCart(productId) {  
  cart = cart.filter((item) => item.id !== productId);  
  saveCart();  
  renderCart();  
}  
  
function updateQuantity(productId, newQty) {  
  newQty = parseInt(newQty, 10);  
  const item = cart.find((i) => i.id === productId);  
  if (!item) return;  
  if (isNaN(newQty) || newQty < 1) newQty = 1;  
  item.quantity = newQty;  
  saveCart();  
  calculateTotal();  
}
```

```

/* ----- 8. Cart badge (every page) ----- */
function updateCartBadge() {
  const badge = document.querySelector("#cart-count-badge");
  if (!badge) return;
  const count = cart.reduce((sum, item) => sum + item.quantity, 0);
  badge.textContent = count;
  badge.style.display = count > 0 ? "inline-block" : "none";
}

/* ----- 2 & 3. Render products on products.html ----- */
function renderProducts() {
  const grid = document.querySelector("#product-grid");
  if (!grid) return;

  grid.innerHTML = products
    .map(
      (p) => `
      <div class="product-card">
        
        <h3>${p.name}</h3>
        <p>₹${p.price.toLocaleString("en-IN")}</p>
        <button class="add-to-cart-btn" data-id="${p.id}">Add to Cart</button>
      </div>`
    )
    .join("");

  grid.querySelectorAll(".add-to-cart-btn").forEach((btn) => {
    btn.addEventListener("click", (e) => {
      const id = parseInt(e.target.dataset.id, 10);
      addToCart(id);
      e.target.textContent = "Added ✓";
      e.target.classList.add("added");
      setTimeout(() => {
        e.target.textContent = "Add to Cart";
        e.target.classList.remove("added");
      }, 900);
    });
  });
}

/* ----- 4 & 5. Render cart on cart.html ----- */
function renderCart() {
  const cartBody = document.querySelector("#cart-body");
  const cartTable = document.querySelector("#cart-table");
  const emptyMsg = document.querySelector("#empty-cart-msg");
  if (!cartBody) return;

  if (cart.length === 0) {
    if (cartTable) cartTable.style.display = "none";
    if (emptyMsg) emptyMsg.style.display = "block";
    calculateTotal();
    return;
  }

  if (cartTable) cartTable.style.display = "table";
  if (emptyMsg) emptyMsg.style.display = "none";

  cartBody.innerHTML = cart
    .map(
      (item) => `
      <tr data-id="${item.id}">
        <td>${item.name}</td>
        <td>
          <input type="number" class="qty-input" min="1" value="${item.quantity}" data-id="${
item.id}">
        </td>
        <td>₹${item.price.toLocaleString("en-IN")}</td>
        <td class="subtotal">₹${(item.price * item.quantity).toLocaleString("en-IN")}</td>
      </tr>`
    )
    .join("");
}

```

```

        <td><button class="remove-btn" data-id="${item.id}">Remove</button></td>
    </tr>`
    )
    .join("");

    cartBody.querySelectorAll(".qty-input").forEach((input) => {
        input.addEventListener("change", (e) => {
            const id = parseInt(e.target.dataset.id, 10);
            updateQuantity(id, e.target.value);
            renderCart();
        });
    });

    cartBody.querySelectorAll(".remove-btn").forEach((btn) => {
        btn.addEventListener("click", (e) => {
            const id = parseInt(e.target.dataset.id, 10);
            removeFromCart(id);
        });
    });

    calculateTotal();
}

/* ----- 5. calculateTotal() ----- */
function calculateTotal() {
    const totalEl = document.querySelector("#grand-total");
    if (!totalEl) return;
    const total = cart.reduce((sum, item) => sum + item.price * item.quantity, 0);
    totalEl.textContent = "\u20B9" + total.toLocaleString("en-IN");
    return total;
}

/* ----- 6 & 7. Checkout form validation ----- */
function initCheckoutForm() {
    const form = document.querySelector("#checkout-form");
    if (!form) return;

    form.addEventListener("submit", (e) => {
        e.preventDefault();
        let isValid = true;

        const fields = [
            { id: "name", errorId: "name-error", validate: (v) => v.trim() !== "", message: "Name is required." },
            { id: "address", errorId: "address-error", validate: (v) => v.trim() !== "", message: "Address is required." },
            { id: "pincode", errorId: "pincode-error", validate: (v) => /^[0-9]{6}$/.test(v.trim()), message: "Pincode must be exactly 6 digits." },
            { id: "phone", errorId: "phone-error", validate: (v) => /^[0-9]{10}$/.test(v.trim()), message: "Phone number must be exactly 10 digits." }
        ];

        fields.forEach((field) => {
            const input = document.querySelector("#" + field.id);
            const errorSpan = document.querySelector("#" + field.errorId);
            const valid = field.validate(input.value);

            if (!valid) {
                isValid = false;
                input.classList.add("error-input");
                if (errorSpan) errorSpan.textContent = field.message;
            } else {
                input.classList.remove("error-input");
                if (errorSpan) errorSpan.textContent = "";
            }
        });

        if (isValid) {

```

```
    form.style.display = "none";
    const confirmationMsg = document.querySelector("#confirmation-msg");
    if (confirmationMsg) confirmationMsg.style.display = "block";

    localStorage.removeItem("cart");
    cart = [];
    updateCartBadge();
  }
});
}

/* ----- Init on every page load ----- */
document.addEventListener("DOMContentLoaded", () => {
  updateCartBadge();
  renderProducts();
  renderCart();
  initCheckoutForm();
});
```

PART: 02 UPDATED HTML FILES

1. INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TechMart - Home</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="navbar">
    <div class="logo">TechMart</div>
    <nav>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart <span id="cart-count-badge"
class="cart-badge">0</span></a></li>
      </ul>
    </nav>
  </div>
</header>

<section class="hero">
  <h1>Welcome to TechMart</h1>
  <p>Your one-stop shop for the latest gadgets and accessories.</p>
  <a href="products.html"><button>Shop Now</button></a>
</section>

<section>
  <h2>Featured Products</h2>
  <div class="product-grid">
    <div class="product-card">
      
      <h3>Wireless Earbuds</h3>
      <p>₹1,999</p>
    </div>
    <div class="product-card">
      
      <h3>Smart Watch</h3>
      <p>₹3,499</p>
    </div>
    <div class="product-card">
      
      <h3>Bluetooth Speaker</h3>
      <p>₹1,299</p>
    </div>
    <div class="product-card">
      
      <h3>Power Bank</h3>
      <p>₹999</p>
    </div>
  </div>
</section>

<footer>
  <p>Contact us: support@techmart.com | +91 98765 43210</p>
  <p>&copy; 2026 TechMart. All rights reserved.</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```


2. PRODUCTS.HTML

The hardcoded product cards from Lab Sheet 1 were replaced with a single empty `#product-grid` container. `script.js` renders all 8 cards from the `products[]` array using template literals.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TechMart - Products</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="navbar">
    <div class="logo">TechMart</div>
    <nav>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart <span id="cart-count-badge"
class="cart-badge">0</span></a></li>
      </ul>
    </nav>
  </div>
</header>

<section>
  <h2>All Products</h2>
  <!-- Product cards are rendered dynamically by script.js from the products[] array -->
  <div class="product-grid" id="product-grid"></div>
</section>

<footer>
  <p>Contact us: support@techmart.com | +91 98765 43210</p>
  <p>&copy; 2026 TechMart. All rights reserved.</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

3. CART.HTML

#cart-body is populated dynamically from localStorage. Each row has a quantity <input> and a Remove button wired up by script.js.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TechMart - Cart</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="navbar">
    <div class="logo">TechMart</div>
    <nav>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart <span id="cart-count-badge"
class="cart-badge">0</span></a></li>
      </ul>
    </nav>
  </div>
</header>

<section>
  <h2>Your Cart</h2>

  <p id="empty-cart-msg" class="empty-cart-msg" style="display:none;">
    Your cart is empty. <a href="products.html">Continue shopping</a>.
  </p>

  <table id="cart-table">
    <tr>
      <th>Item</th>
      <th>Quantity</th>
      <th>Price</th>
      <th>Subtotal</th>
      <th>Action</th>
    </tr>
    <!-- Cart rows are rendered dynamically by script.js from localStorage -->
    <tbody id="cart-body"></tbody>
    <tr class="total-row">
      <td colspan="3">Grand Total</td>
      <td id="grand-total">₹0</td>
      <td></td>
    </tr>
  </table>

  <div style="text-align:center;">
    <a href="checkout.html"><button style="padding:10px 20px;background:var(--accent-
color);border:none;border-radius:6px;color:#fff;">Proceed to Checkout</button></a>
  </div>
</section>

<footer>
  <p>Contact us: support@techmart.com | +91 98765 43210</p>
  <p>&copy; 2026 TechMart. All rights reserved.</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

4. CHECKOUT.HTML

Every required field has an inline `<span class="error-msg">` next to it. No `alert()` is used anywhere — validation messages are written directly into these spans.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TechMart - Checkout</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="navbar">
    <div class="logo">TechMart</div>
    <nav>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart <span id="cart-count-badge"
class="cart-badge">0</span></a></li>
      </ul>
    </nav>
  </div>
</header>

<section>
  <h2>Checkout</h2>

  <form id="checkout-form" novalidate>
    <label for="name">Full Name</label>
    <input type="text" id="name" name="name">
    <span class="error-msg" id="name-error"></span>

    <label for="address">Address</label>
    <input type="text" id="address" name="address">
    <span class="error-msg" id="address-error"></span>

    <label for="city">City</label>
    <input type="text" id="city" name="city">

    <label for="pincode">Pincode</label>
    <input type="text" id="pincode" name="pincode" maxlength="6">
    <span class="error-msg" id="pincode-error"></span>

    <label for="phone">Phone</label>
    <input type="tel" id="phone" name="phone" maxlength="10">
    <span class="error-msg" id="phone-error"></span>

    <label>Payment Method</label>
    <div class="payment-options">
      <label><input type="radio" name="payment" value="cod" checked> Cash on Delivery</label>
      <label><input type="radio" name="payment" value="upi"> UPI</label>
      <label><input type="radio" name="payment" value="card"> Card</label>
    </div>

    <button type="submit">Place Order</button>
  </form>

  <p id="confirmation-msg" class="confirmation-msg" style="display:none;">
    🟢 Order placed! Thank you for shopping with TechMart.
  </p>
</section>

<footer>
  <p>Contact us: support@techmart.com | +91 98765 43210</p>
  <p>&copy; 2026 TechMart. All rights reserved.</p>
</footer>
```

```
</footer>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

5. PRODUCT-DETAIL.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>TechMart - Product Detail</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="navbar">
    <div class="logo">TechMart</div>
    <nav>
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart <span id="cart-count-badge"
class="cart-badge">0</span></a></li>
      </ul>
    </nav>
  </div>
</header>

<section class="product-detail">
  
  <div class="info">
    <h2>Smart Watch</h2>
    <p>A feature-packed smart watch with heart-rate monitoring, sleep tracking,
    and a 7-day battery life. Water resistant and compatible with Android & iOS.</p>
    <p><strong>Price: ₹3,499</strong></p>
    <label for="qty">Quantity:</label>
    <input type="number" id="qty" name="qty" value="1" min="1" style="width:60px;">
    <br>
    <button id="detail-add-btn" data-id="2">Add to Cart</button>
  </div>
</section>

<footer>
  <p>Contact us: support@techmart.com | +91 98765 43210</p>
  <p>&copy; 2026 TechMart. All rights reserved.</p>
</footer>

<script src="script.js"></script>
<script>
  document.querySelector("#detail-add-btn").addEventListener("click", () => {
    const qty = parseInt(document.querySelector("#qty").value, 10) || 1;
    for (let i = 0; i < qty; i++) addToCart(2);
    alert("Added to cart!");
  });
</script>
</body>
</html>
```

6. STYLE.CSS (additions for Lab Sheet 2)

Base styling carried over from Lab Sheet 1; new rules added for the cart badge, quantity input, remove button, inline error messages and the order confirmation banner.

```
:root {
  --primary-color: #1f2937;
  --accent-color: #f97316;
  --bg-color: #f9fafb;
  --card-bg: #ffffff;
  --text-muted: #6b7280;
}
* { box-sizing: border-box; }
body {
  margin: 0;
  font-family: Arial, Helvetica, sans-serif;
  background: var(--bg-color);
  color: var(--primary-color);
}
/* ----- Header / Navbar ----- */
header {
  background: var(--primary-color);
  color: #fff;
  padding: 14px 24px;
}
.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
}
.logo { font-size: 1.4rem; font-weight: bold; }
nav ul {
  list-style: none;
  display: flex;
  gap: 20px;
  margin: 0;
  padding: 0;
}
nav a {
  color: #fff;
  text-decoration: none;
  position: relative;
}
nav a:hover { color: var(--accent-color); }
.cart-badge {
  background: var(--accent-color);
  color: #fff;
  font-size: 0.7rem;
  font-weight: bold;
  border-radius: 999px;
  padding: 1px 7px;
  margin-left: 6px;
}
/* ----- Hero ----- */
.hero {
  background: linear-gradient(135deg, var(--primary-color), #374151);
  color: #fff;
  text-align: center;
  padding: 60px 20px;
}
.hero button {
  margin-top: 16px;
  padding: 10px 24px;
  background: var(--accent-color);
  border: none;
  border-radius: 6px;
  color: #fff;
}
```

```

    cursor: pointer;
}
section h2 { text-align: center; }
/* ----- Product grid & cards ----- */
.product-grid {
  display: flex;
  flex-wrap: wrap;
  gap: 20px;
  padding: 30px;
  justify-content: center;
}
.product-card {
  background: var(--card-bg);
  border-radius: 10px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
  padding: 16px;
  width: 220px;
  text-align: center;
}
.product-card img { width: 100%; height: 160px; object-fit: contain; background: #fff; border-
radius: 6px; }
.product-card button {
  margin-top: 10px;
  padding: 8px 14px;
  background: var(--accent-color);
  border: none;
  border-radius: 6px;
  color: #fff;
  cursor: pointer;
}
.product-card button:hover { background: #ea580c; }
.product-card button.added { background: #16a34a; }
/* ----- Product detail ----- */
.product-detail {
  display: flex;
  gap: 30px;
  padding: 30px;
  flex-wrap: wrap;
}
.product-detail img { width: 320px; height: 320px; object-fit: contain; background: #fff;
border: 1px solid #e5e7eb; border-radius: 10px; }
.product-detail .info { max-width: 400px; }
/* ----- Cart table ----- */
table {
  width: 90%;
  margin: 30px auto;
  border-collapse: collapse;
  background: var(--card-bg);
}
table th, table td {
  border: 1px solid #e5e7eb;
  padding: 10px;
  text-align: center;
}
table th { background: var(--primary-color); color: #fff; }
.total-row td { font-weight: bold; background: #f3f4f6; }
.qty-input { width: 55px; padding: 4px; text-align: center; }
.remove-btn {
  background: #dc2626;
  border: none;
  color: #fff;
  padding: 6px 12px;
  border-radius: 6px;
  cursor: pointer;
}
.remove-btn:hover { background: #b91c1c; }
.empty-cart-msg { text-align: center; color: var(--text-muted); padding: 30px; }
/* ----- Checkout form ----- */

```

```

form {
  max-width: 500px;
  margin: 30px auto;
  background: var(--card-bg);
  padding: 24px;
  border-radius: 10px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}
form label { display: block; margin-top: 12px; margin-bottom: 4px; }
form input[type="text"],
form input[type="tel"] {
  width: 100%;
  padding: 8px;
  border: 1px solid #d1d5db;
  border-radius: 6px;
}
form input:focus { outline: 2px solid var(--accent-color); border-color: var(--accent-color); }
form input.error-input { border-color: #dc2626; }
.error-msg {
  color: #dc2626;
  font-size: 0.82rem;
  margin-top: 4px;
  min-height: 1em;
}
.payment-options label { display: inline-block; margin-right: 16px; }
form button {
  margin-top: 20px;
  padding: 10px 20px;
  background: var(--accent-color);
  border: none;
  border-radius: 6px;
  color: #fff;
  cursor: pointer;
}
form button:hover { background: #ea580c; }
.confirmation-msg {
  text-align: center;
  background: #dcfce7;
  color: #166534;
  border: 1px solid #86efac;
  padding: 16px;
  border-radius: 8px;
  margin-top: 16px;
  font-weight: bold;
}
/* ----- Footer ----- */
footer {
  background: var(--primary-color);
  color: #d1d5db;
  text-align: center;
  padding: 16px;
  margin-top: 30px;
}
/* ----- Responsive ----- */
@media (max-width: 600px) {
  .navbar { flex-direction: column; align-items: flex-start; }
  nav ul { flex-direction: column; gap: 8px; margin-top: 10px; }
  .product-grid { flex-direction: column; align-items: center; }
  .product-detail { flex-direction: column; }
  table { width: 100%; font-size: 0.9rem; }
}

```

OUTPUT SCREENSHOTS

products.html — dynamically rendered from the products[] array

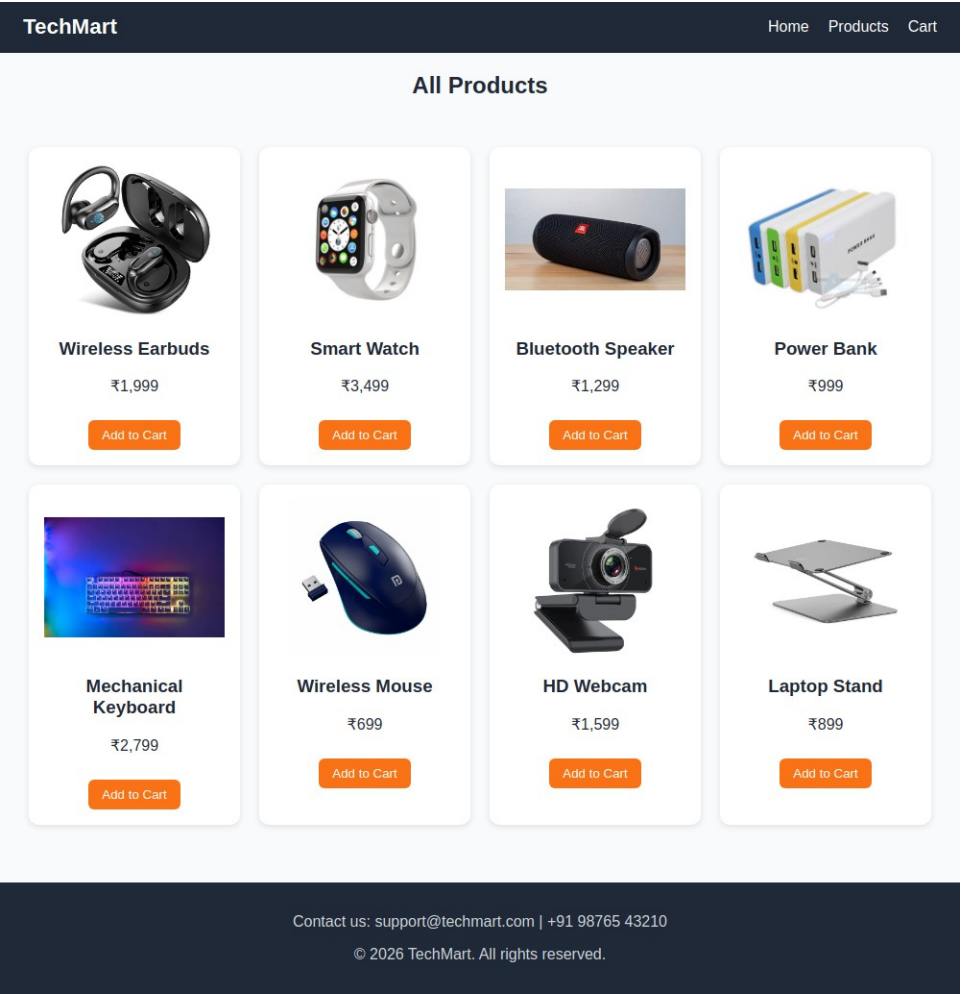


Fig 1: All 8 product cards generated by renderProducts() using createElement/innerHTML with template literals.

Add to Cart in action — cart badge updates live

TechMart

HomeProductsCart4

All Products



Wireless Earbuds

₹1,999

Added ✓



Smart Watch

₹3,499

Added ✓



Bluetooth Speaker

₹1,299

Add to Cart



Power Bank

₹999

Added ✓



Mechanical Keyboard

₹2,799

Add to Cart



Wireless Mouse

₹699

Add to Cart



HD Webcam

₹1,599

Add to Cart



Laptop Stand

₹899

Add to Cart

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Fig 2: Earbuds (x2), Smart Watch and Power Bank added. Navbar badge shows 4 items, read live from localStorage.

cart.html — items rendered from localStorage

TechMart

[Home](#) [Products](#) [Cart](#)

4

Your Cart

Item	Quantity	Price	Subtotal	Action
Wireless Earbuds	2	₹1,999	₹3,998	<button>Remove</button>
Smart Watch	1	₹3,499	₹3,499	<button>Remove</button>
Power Bank	1	₹999	₹999	<button>Remove</button>
Grand Total			₹8,496	

Proceed to Checkout

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Fig 3: Cart page rendered from localStorage with quantity inputs, subtotal and grand total.

Quantity change recalculates the grand total instantly

TechMart

[Home](#) [Products](#) [Cart](#)

5

Your Cart

Item	Quantity	Price	Subtotal	Action
Wireless Earbuds	3	₹1,999	₹5,997	<button>Remove</button>
Smart Watch	1	₹3,499	₹3,499	<button>Remove</button>
Power Bank	1	₹999	₹999	<button>Remove</button>
Grand Total			₹10,495	

Proceed to Checkout

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Fig 4: Earbuds quantity changed to 3 → subtotal and Grand Total (₹10,495) recalculated by calculateTotal().

checkout.html — inline validation errors (no alert())

TechMart

HomeProductsCart0

Checkout

Full Name

Name is required.

Address

Address is required.

City

Pincode

123

Pincode must be exactly 6 digits.

Phone

98765

Phone number must be exactly 10 digits.

Payment Method

☒ Cash on Delivery☐ UPI☐ Card

Place Order

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Fig 5: Empty Name/Address, 3-digit Pincode and 5-digit Phone all rejected with inline red error text under each field.

Successful order placement

TechMart

HomeProductsCart0

Checkout

✔ Order placed! Thank you for shopping with TechMart.

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Fig 6: Valid data submitted → confirmation message shown, cart badge reset to 0, localStorage cart cleared.

Cart is empty after order is placed

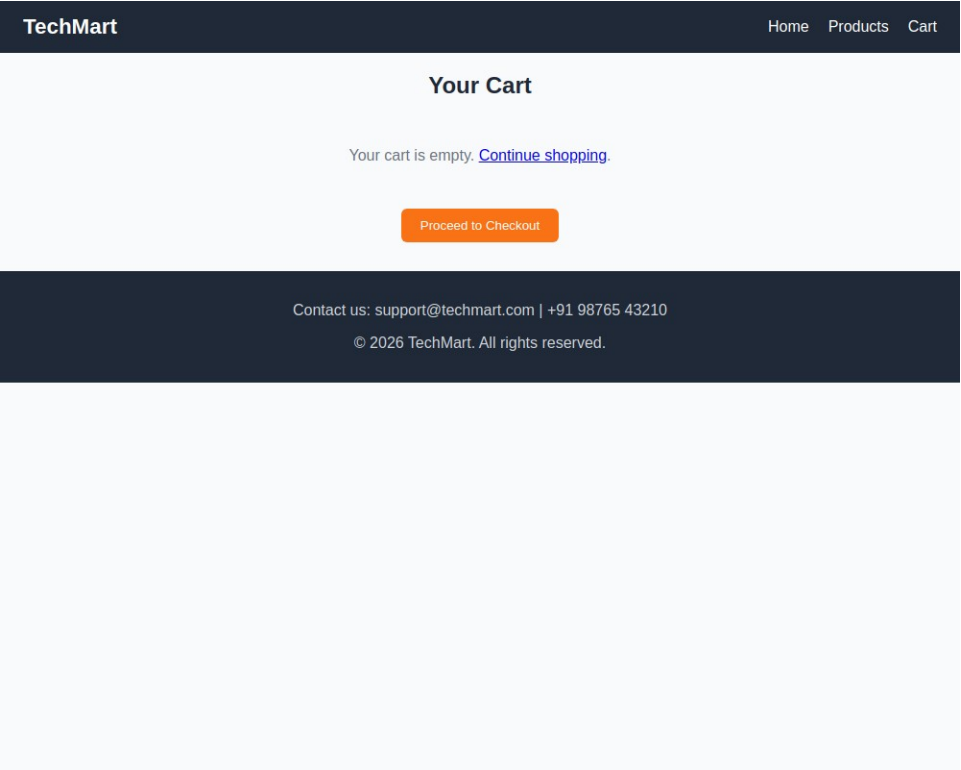


Fig 7: Navigating back to cart.html after checkout confirms localStorage was cleared as required by Task 7.